Compile time

User rule (matrix example)

```
(dot[@d]
  (transpose[@tr] A) B)
=> (dot[auto] A B)
```

auto: generate attributes

Operation semantics

```
transpose
dot
```

TEPL compiler

Synthesize + verify

Generated attribute function

Generated once for this rule

```
f(p, q) → replacement dot attrs
p = @tr.permutation
q = @d
```

Runtime

Evaluate on each match

Captured attrs

```
p = [1, 0]
```

q: left axis = 0

Apply f

```
f(p, q)
```

New dot attrs

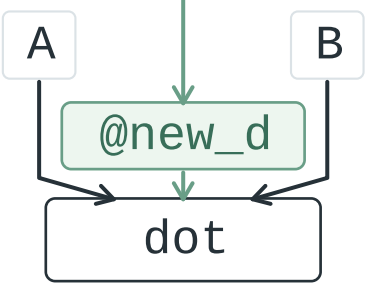
left axis = 1

@new_d

Apply replacement

```
(dot[@new_d] A B)
```

Transpose removed



Left axis = left contraction axis.
Only the changing dot field is shown.